

Jinwei Liu

Ph.D. Student in Control Science and Engineering

📍 Hefei, China ✉ liujinweishandong@outlook.com 🌐 jinwei-liu.github.io 👤 Jinwei-Liu

Profile

- Ph.D. student in Control Science and Engineering at the University of Science and Technology of China through a master's-to-Ph.D. transfer, advised by Prof. Yun-Bo Zhao in the Networked Intelligent Control Lab (NICLab).
- During my master's study, I worked with Pengfei Li, Special Associate Researcher at USTC.
- Research interests include human-machine hybrid intelligence, human-robot interaction, shared autonomy, and embodied intelligence.
- Current work focuses on shared control, teleoperation, reinforcement learning, and human-in-the-loop reinforcement learning.
- Interested in both human-to-machine intent communication and machine-to-human information expression.

Experience

Dobot, Embodied Intelligence Algorithm Intern Shanghai, China
June 2026 – present
1 month

Education

Doc- University of Science and Technology of China, Control Science and Engineering Hefei, China
tor of • Master's-to-Ph.D. transfer; combined master's-doctoral track June 2026 – present
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Mas- University of Science and Technology of China, Artificial Intelligence Hefei, China
ter of • Weighted average: 91.09; GPA: 3.99/4.3 Sept 2024 – June 2026
Engi-
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Bach- Ocean University of China, Automation Qingdao, China
elor of • Rank: 1/89; Weighted average: 92.42; GPA: 3.82/4.0 Sept 2020 – June 2024
Engi- • Honors: National Scholarship (twice); Honorary Bachelor's Degree of Ocean Uni-
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Publications

Delay-Aware Shared Control for Teleoperation Systems: Intent Trajectory Prediction 2026
Under Communication Latency
Journal article on delay-aware shared control for teleoperation systems, using DA-RT to predict operator intent under communication latency.
Jinwei Liu, Pengfei Li, Yaqing Zhou, Tao Wang, Yu Kang, Yun-Bo Zhao
ieeexplore.ieee.org/document/11535330

Legible Shared Autonomy: Implicit Communication of Robot Belief through Motion Accepted paper on legible shared autonomy, enabling robot assistive motion to communicate inferred user goals and confidence through motion. Jinwei Liu, Pengfei Li, Shaofeng Chen, Tao Wang, Yun-Bo Zhao	2026
LSTM-MPC: Physically Feasible and Interpretable Trajectory Prediction Accepted paper on physically feasible and interpretable trajectory prediction via latent MPC parameter estimation. Jinwei Liu, Pengfei Li, Qianqian Zhang, Yunbo Zhao, Yu Kang	2026
Human-AI Shared Control via Motor Knowledge Neuron Stimulation Accepted paper on human-AI shared control through distributed motor knowledge neuron assemblies. Jinwei Liu, Pengfei Li, Sibin Huang, Yun-Bo Zhao, Yu Kang	2026